

CHEATSHEET · QUICK REFERENCE

Lab 16 — Scripting Basics for Server Administration

Write bash with variables, data types, conditionals, comparators and loops, then automate service checks, bulk account creation and host bootstrap.

Script types (memorise)

Type	OS	Extension
Bash	Linux/Unix	.sh
Batch	Windows	.bat / .cmd
PowerShell	Windows	.ps1
VBScript	Windows (legacy)	.vbs

Variables, types and env

```
NAME="srv01"
Local variable

export ADMIN_EMAIL="ops@lab.local"
Environment variable (inherited)

COUNT=$((COUNT + 1))
Integer arithmetic

echo "${#GREETING}"
String length

SERVICES=(nginx mariadb sshd)
Array declaration
```

Comparators

Numeric	String	Meaning
-eq	==	equal
-ne	!=	not equal
-lt	<	less than
-gt	>	greater than
-ge	>=	greater or equal

Admin task patterns

```
for s in "${SERVICES[@]}"; do systemctl is-active "$s"; done
Service status loop

id "$user" || useradd -m -c "$fullname" -G "$group" "$user"
Idempotent account creation

set -euo pipefail
Production bootstrap discipline
```